

9 (short)

(a) Explain briefly, using sketches to support your answer, the microstructural origins of the Young's modulus and the yield strength of a pure metal. Would you expect both to change after cold rolling? [6]

(b) Pure Cu and pure Ni have yield strengths of 60 MPa and 80 MPa, respectively. Alloys of Cu and Ni form a solid solution at room temperature, without precipitation, for any composition. Explain, with reference to the relevant hardening mechanisms, which of the following alloys (expressed in number fraction of atoms) will have the highest yield strength, measured after casting: (i) Cu - 10% Ni, (ii) Cu - 50% Ni, (iii) Cu - 90% Ni. [4]